

Crashworthiness Evaluation of 3D-Printed Origami-Inspired PLA-CF Crash Boxes

Heshaam Abdullah Alotaibi (77675), Mohammed Altemeemi (66220), Yousef Ghanim Abadi (56318)Sari M. S. Almutairi (40744)Meshal M. Almesaieed (59192)

Supervisor Name: Dr. Mehmet Guler

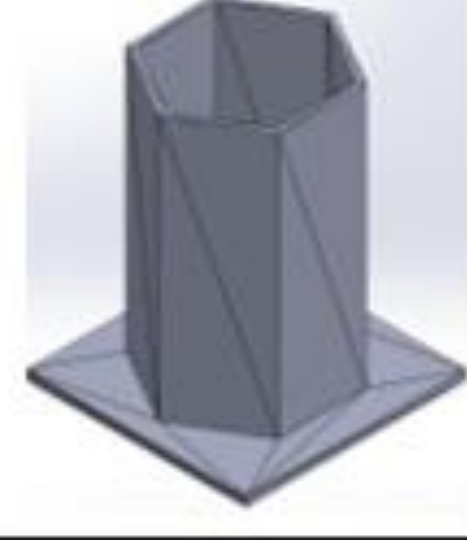
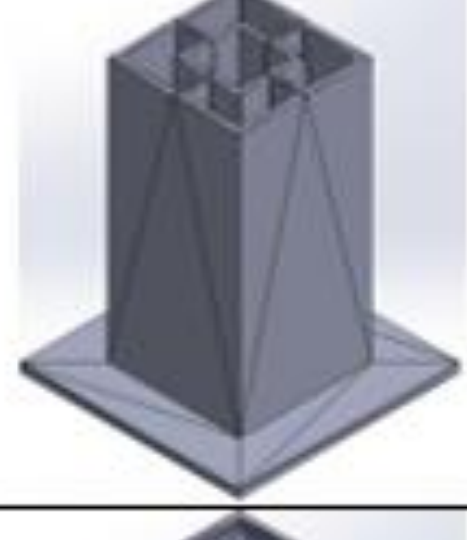
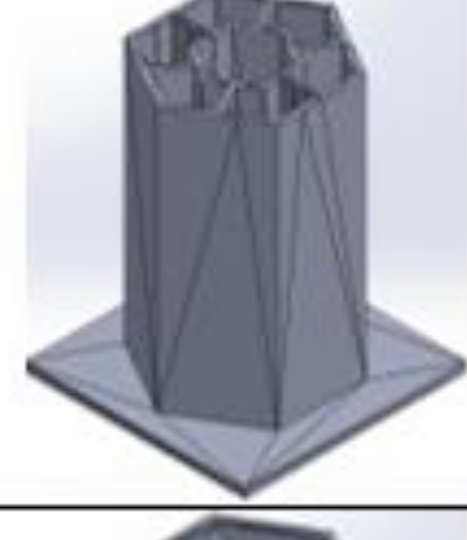
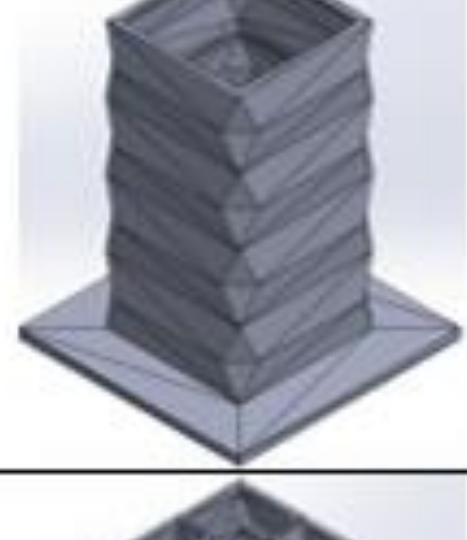
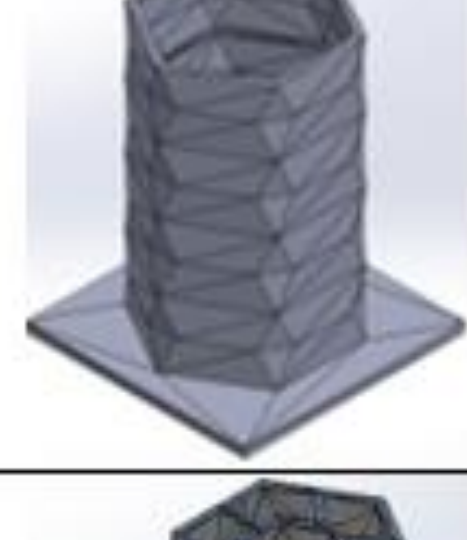
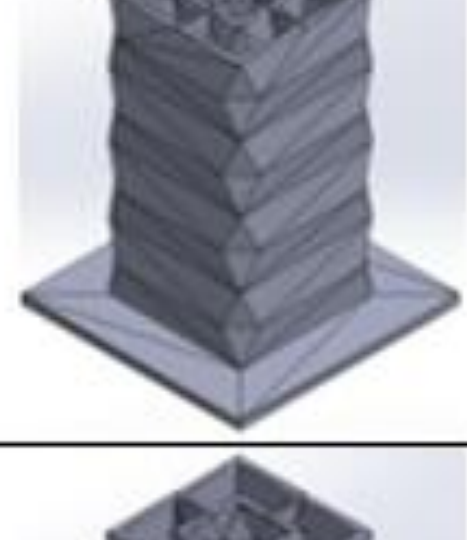
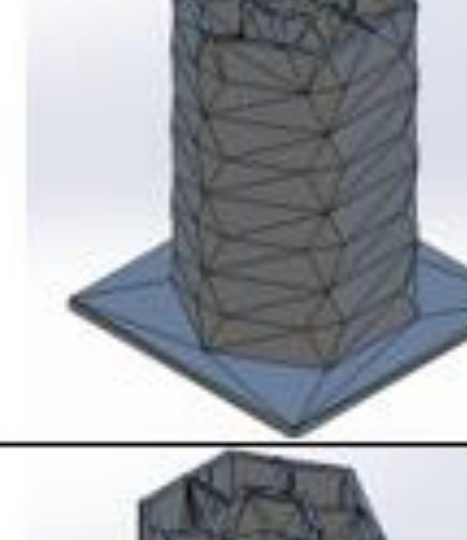
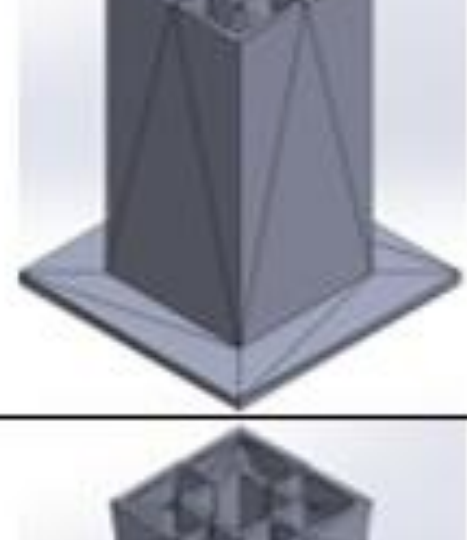
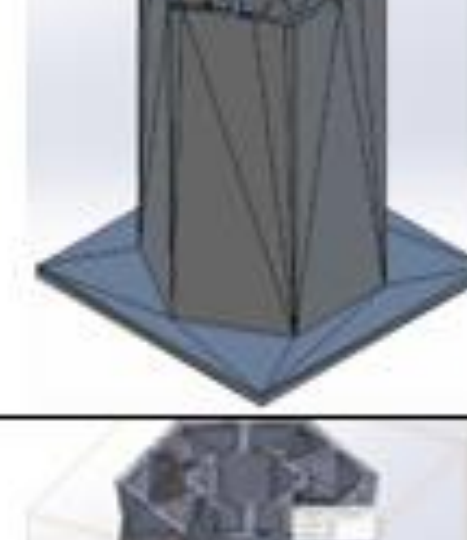
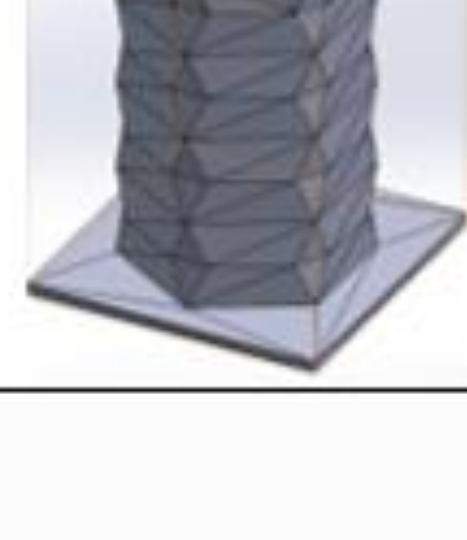
PROBLEM STATEMENT, CRITERIA & CONSTRAINTS

- Problem Statement
- Need for lightweight, efficient crash absorbers.
 - Traditional crash boxes: Heavy, Expensive and Complex to manufacture.
 - Lack of research on 3D-printed polymer crash structures
- Evaluation Criteria
- Specific Energy Absorption (SEA)
 - Peak Crush Force (PCF)
 - Mean Crush Force (MCF)
 - Crash Force Efficiency (CFE)
 - Structural stability
- Constraints
- Material limits (PLA-CF only)
 - 3D printing defects & accuracy
 - Testing limited to ~15 km/h
 - Equipment limitations

EXISTING SOLUTION & LIMITATIONS

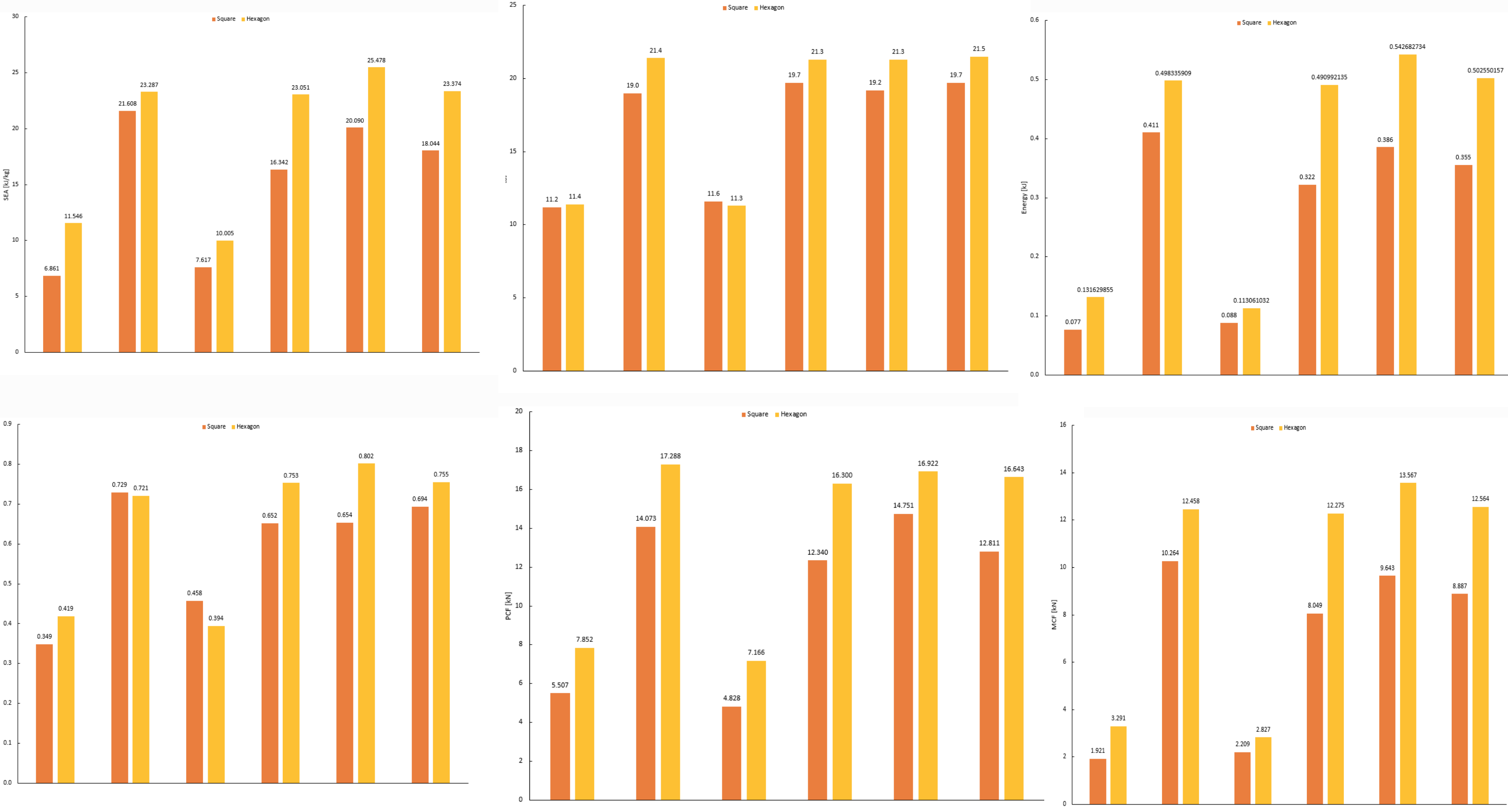
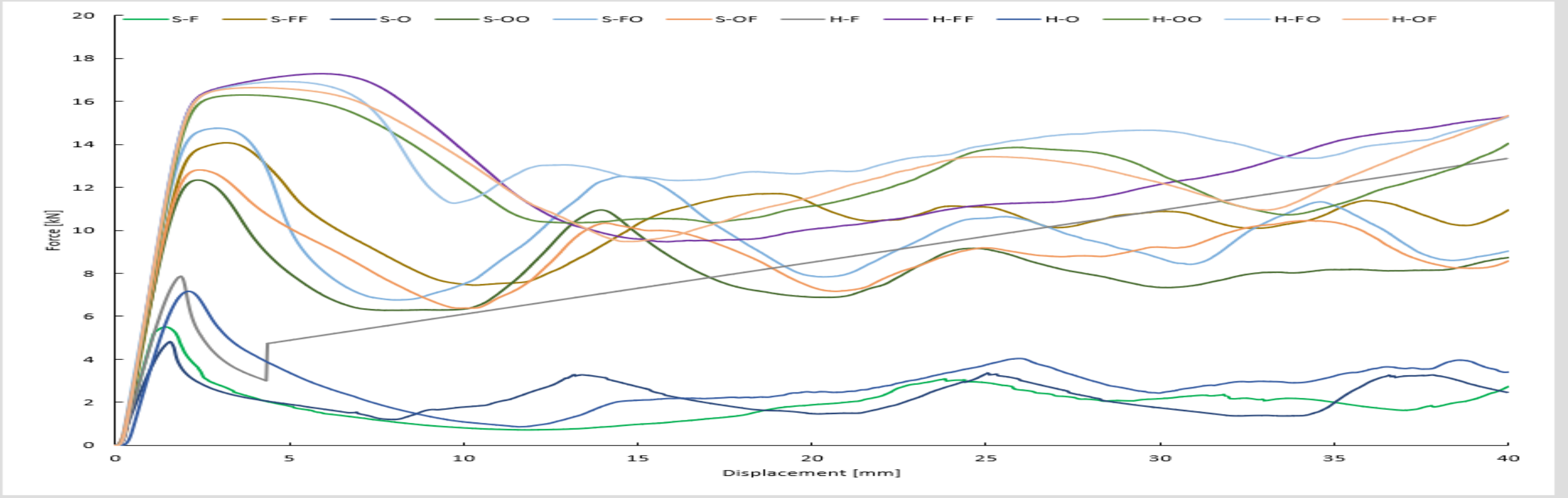
- Existing Solutions
- Metallic Crash-boxes
 - Composite Structures(CFRP, GFRP)
 - 3D-Printed Polymer Structures (PLA, PLA-CF)
- Limitations
- High weight in metallic structures → reduced fuel efficiency
 - High manufacturing and material costs (metals & composites)
 - Limited design flexibility using traditional manufacturing methods
 - Complex and expensive fabrication for composite materials
 - Brittle failure behavior in composites
 - Lower strength and anisotropic behavior in 3D-printed polymers
 - Sensitivity to printing parameters and defects (FDM limitations)
 - Limited validation under real crash conditions
- Research Gaps
- Insufficient data on crashworthiness of 3D-printed polymers (PLA-CF).
 - Limited studies combining:
 - Material + geometry optimization together
 - Lack of research on origami-inspired folding structures in crash applications.
 - Few studies comparing different cross-sectional shapes (square vs hexagonal).
 - Limited experimental validation alongside FEA simulations.
 - Need for optimized designs that balance: Energy absorption, Peak force reduction and Structural stability.

PROPOSED SOLUTION & DESIGN

	SQUARE		HEXAGON
S-F		H-F	
S-FF		H-FF	
S-O		H-O	
S-OO		H-OO	
S-FO		H-FO	
S-OF		H-OF	

IMPLEMENTATION & RESULTS

Configuration	Mass [g]	Energy [kJ]	PCF [kN]	MCF [kN]	SEA [kJ/kg]	CFE
S-F	11.2	0.077	5.507	1.921	6.861	0.349
S-FF	19.0	0.411	14.073	10.264	21.608	0.729
S-O	11.6	0.088	4.828	2.209	7.617	0.458
S-OO	19.7	0.322	12.340	8.049	16.342	0.652
S-FO	19.2	0.386	14.751	9.643	20.090	0.654
S-OF	19.7	0.355	12.811	8.887	18.044	0.694
H-F	11.4	0.132	7.852	3.291	11.546	0.419
H-FF	21.4	0.498	17.288	12.458	23.287	0.721
H-O	11.3	0.113	7.166	2.827	10.005	0.394
H-OO	21.3	0.491	16.300	12.275	23.051	0.753
H-FO	21.3	0.543	16.922	13.567	25.478	0.802
H-OF	21.5	0.503	16.643	12.564	23.374	0.755



CONCLUSIONS & FUTURE WORK

This study demonstrates that 3D-printed PLA-CF crash boxes with origami-inspired geometries can provide a lightweight, sustainable, and high-performance alternative to conventional metallic structures, with the hexagonal flat-origami (H-FO) configuration achieving the best energy absorption and stability. The results confirm that combining geometry optimization with material selection significantly enhances crashworthiness by enabling controlled deformation and reducing peak forces. Future work will focus on evaluating performance under higher impact speeds, investigating environmental effects, and exploring multi-material printing to further improve structural efficiency and real-world applicability.

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